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Emotion induction by Western music across personality types using Internet of Things Technology

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Abstract: The traditional music recommendation algorithm is an algorithm that measures the similarity of user preferences, or an algorithm that distinguishes types according to music styles and genres, which cannot meet the needs of emotional induction for different personality types. It is necessary to study the emotional induction of varying personality types by Western music based on the Internet of Things (IoT) technology. Electroencephalogram (EEG) emotion recognition based on IoT technology is a new research field, which involves emotion induction, EEG feature extraction, and pattern recognition technology. According to the dimensional model of emotion, this paper selects three kinds of Western music fragments, which can express neutral, positive, and negative emotions. It uses these Western music materials to induce an EEG in three emotional states. Comparing the classification effects of the eigenvectors of different rhythms, it is found that the classification with the eigenvectors of the beta and gamma rhythms has the highest accuracy, with the overall average accuracy of 0.842 and 0.841, respectively, and the electrodes that provide features under these two rhythms are in the head. The location distribution of the table was consistent between subjects. Comparing the classification effects of different classifiers, it is found that support vector machine (SVM) and Query-by-Committee (QBC) are better than other classifiers, and the highest average correct rates between subjects on different rhythms are 94.7% and 90.0%, respectively.

Keywords: Emotion Induction, Internet of Things Technology, Western Music, Emotion Feature Learning, EEG

1. Introduction

Most of the common emotion measures are completed by the subjects' subjective emotion label selection [1]. However, subjective scales are affected by aspects such as word comprehension and may not fully express the true experience of emotions. EEG signals are relatively more feasible and accurate for emotion recognition by reflecting the activation state of human brain functions [2]. Research on user emotions has been the focus of multiple disciplines. Numerous

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studies in neuroscience, cognitive science, and biology have shown that emotions play a critical role in rational and intelligent behavior. A person's emotional state not only affects an individual's attention span but also has an impact on the ability to solve problems and make decisions [3]. Finding and calculating user emotions has important practical application value. Adding user emotions when profiling users can make user portraits more accurate, to provide users with better personalized services and improve users' quality of life.

Although virtual reality (VR) is increasingly recognized as an effective emotion-inducing technology, little research has been done on the relationship between the two. Chirico and Gaggioli [4] introduced 50 people to participate in the experiment to compare the emotional type and valence under different conditions. The results of the study showed that there was no significant difference in the emotions evoked by virtual and natural conditions. How do humans distinguish between emotions and non-emotions? Using the discriminative psychophysical MRI sparse sampling paradigm to locate threshold responses to happy and sad acoustic stimuli, Manno et al. [5] found that threshold emotion recognition in Western music exploits fine structural cues. Western musical anhedonia means the acquired and selective loss of Western musical emotion. The study of Masayuki and Satoh [6] found that there were cases in which Western music emotion was preserved even in the presence of impaired Western music perception and recognition, findings consistent with the findings of activation studies using PET and fMRI [7]. However, it is necessary to explain and explain the experimental results on the basis of understanding the characteristics and limitations of the case. Evans [8] conducts empirical research to explore potential connections between the emotional significance attributed to musical stimuli (expressed emotion, or external locus of emotion) and the personal emotional response triggered by listening to music (felt emotion, or internal locus). It is commonly assumed that the relationship between these two emotional loci is positive, implying that the emotional experience elicited while listening to music aligns with the emotion expressed by the music itself. Soulier [9] aimed to study the effect of negative emotion induction in Western music on the lexical spelling performance of children with and without written language impairment. Studies have found that the effects of emotion vary by spelling level and only affect the performance of children with written language impairments. Kabrin et al. [10] I incorporated a new technology for audiovisual induction of states of consciousness. The study aimed to demonstrate that this technique can induce a relaxed state. He conducted two experiments using synchronous fractals and specific configurations of Western music sequences, which showed that a relaxed state could be induced. The study of Vlker [11] investigated the effects of personal Western music and a researcher's pre-selected Western music on inducing sadness and happiness. Results indicated that participants' choice of Western music had a stronger effect on reported mood, with sadness and happiness mainly elicited by contagion and episodic memories associated with Western music. Wang et al. [12] used the V-A model as the emotional perception model, selected about 1000 classic extracts from China and

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the West, and finally extracted about 20 VITs of different data sets and other emotional dimensions. Valence (V) dimension represents the positivity or negativity of an emotion. Emotions with positive valence are associated with positive feelings like happiness, joy, or love, while emotions with negative valence are associated with negative feelings such as sadness, anger, or fear. At the same time, Arousal (A) refers to the intensity or activation level of an emotion. Emotions with high arousal are intense and stimulating, like excitement or anger, while low arousal emotions are more subdued, like calmness or boredom. The results show that the packaging method combining MaxAbsScaler preprocessing and the recursive feature elimination algorithm based on the maximum random tree is the best algorithm. Harmonic change detection function is a universal feature of culture, while spectral flux is a cultural specificity of Chinese classical music. The study also found that the pitch characteristics of Western classical music are more significant, while the loudness and rhythm characteristics of Chinese classical music are less significant. Sentiment analysis methods are used to evaluate general opinions expressed in tweets about entities. Seghouani et al. [13] proposed a new method to determine entity reputation based on the set of events involved in the entity. Also, they proposed a new sampling method driven by a tweet weighting metric to provide better target entity quality and summarization. To sum up, the activation of emotions is due to the involvement of all levels of the nervous system. The initiation and occurrence of emotions are the result of the integration of nervous systems at all levels and are the product of multiple levels of physiology.

The role of the Internet of Things (IoT) in emotional induction is becoming increasingly important. Tallapragada et al. [14] integrate data intelligence through IoT to track customer sentiment and provide customer behavioral insights. The proposed system uses model-based face and emotion tracking under real-use case conditions. Wu and Zhang [15] carried out a detailed analysis and research on the necessity, feasibility, and implementation of intelligent home voice emotion recognition technology, introduced the definition and classification of emotion, and proposed five primary emotions to be recognized for voice emotion recognition based on the intelligent home environment. Then, on this basis, we analyze the methods of obtaining emotional voice data. Building upon this foundation, the study delves into key aspects of voice data acquisition within smart home environments. It addresses issues like voice characteristics, acquisition methods, and more. Additionally, the study proposes three fundamental principles for voice text design and identifies a suitable hybrid voice recording method tailored for smart homes. Furthermore, the study provides a comprehensive account of the design and establishment process of an emotional voice database for smart homes, offering insights into feature extraction challenges in speech emotion recognition. In the realm of Western music mood prediction, the study focuses on constructing regression models to forecast mood dimensions such as valence (happiness level) and arousal (energy level). Hu and Yang [16] evaluated a mood regression model built on fifteen acoustic features of five mood-related

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aspects of Western music, with a focus on generalization across datasets. Emotion is considered a physiological state that occurs whenever an individual observes a transition in their environment or body. EEG rhythms, such as delta, theta, alpha, beta, and gamma bands, have been widely studied for their role in decoding emotional states, as they reflect different aspects of brain activity associated with affective processing. Prior research has demonstrated that rhythm-based analysis can enhance the accuracy of emotion recognition by capturing frequency-specific neural patterns linked to various affective dimensions. Vempati et al. [17] proposed an EEG rhythm–based emotion recognition approach using multivariate decomposition and an ensemble machine learning classifier, achieving significant improvements in classification performance by leveraging rhythm-specific features. Such studies underscore the relevance of EEG rhythm analysis in emotion recognition tasks and provide strong motivation for incorporating rhythm-based features into IoT-enabled systems for music recommendation. Garg et al. [18] have conducted extensive experiments on different Western music emotion datasets and human emotion for impactful feature extraction, training, testing, and performance evaluation. The above research shows that the role of the Internet of Things in emotional induction should be further explored.

A research gap exists in the current body of studies related to emotion recognition, as there is a limited exploration of cross-cultural emotional recognition. The existing research predominantly focuses on Western contexts, neglecting the potential variations in emotion recognition across different cultures. Investigating the adaptability of IoT-based emotion recognition systems to diverse cultural contexts and understanding how cultural nuances influence emotion recognition is essential for achieving more accurate and globally applicable emotion recognition technology.

In this experiment, we selected 9 Western music clips as experimental stimuli based on their average scores, each clip lasting approximately 11 seconds, and repeated each stimulus 27 times. The experiment had a minimum theoretical duration of 44.5 minutes. Notably, the experimental data revealed that emotional arousal in Western music scenes, including joy, sadness, fear, and disgust, was slightly higher than in videos. Subjective scales offer a straightforward and cost-effective means of gathering individuals' self-reported experiences, making them accessible for large-scale studies and versatile in various fields. However, their reliance on subjective interpretation and potential biases can introduce limitations. On the other hand, an Electroencephalogram (EEG) provides an objective measure of brain activity with high temporal resolution, making it valuable for understanding neural processes. Yet, it has limitations in spatial resolution, sensitivity to artifacts, and complexity in interpretation. The choice between these methods hinges on research goals and practical considerations, with a potential for synergy when used in conjunction to obtain a more comprehensive understanding of both subjective experiences and underlying neural mechanisms. This study's novelty lies in the development of an Internet of Things-based emotion induction system, which evaluates emotions in visual images using image-based

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emotion calculation algorithms. It also contributes to the creation of an intelligent multi-vision fusion annotation model. This research highlights a gap in current studies related to emotion recognition, emphasizing the need to explore cross-cultural emotional recognition and adapt IoT-based systems for diverse cultural contexts, ultimately enhancing the universality and accuracy of emotion recognition technology.

2. Principles and Methods of Emotional Induction

2.1 Principles Related to Emotion Induction

(1) The brain mechanism of emotion

Traditionally, it is believed that information from emotional stimuli is transmitted to the limbic system, which includes the thalamus, hippocampus, and amygdala, from where emotions are generated and expressed. Still, the limbic system is structurally and functionally challenging [19]. The two-loop model of emotion is shown in Figure 1.

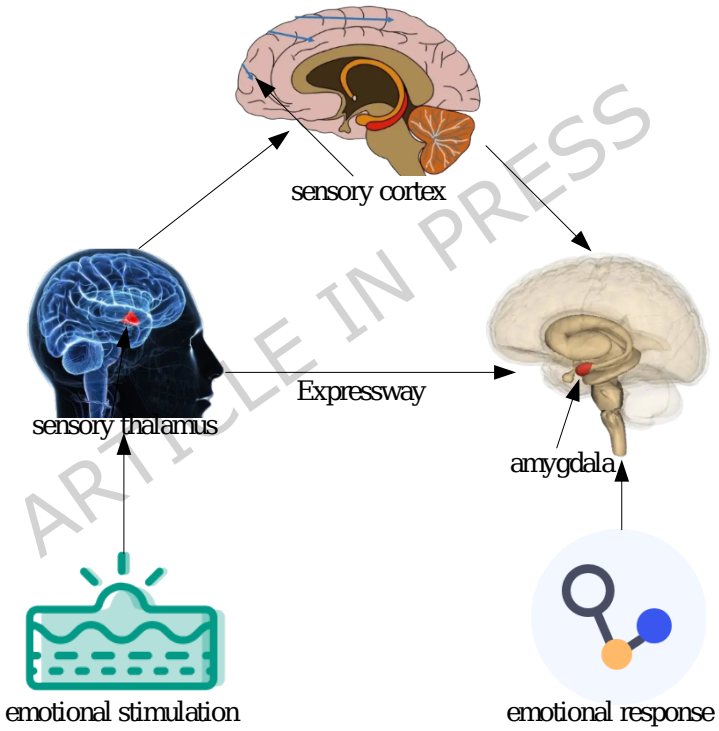

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Figure 1. Two-loop model of emotion

As shown in Figure 1, emotional stimuli are transmitted from the sensory organs through the cortex of the sensory thalamus to the amygdala, which immediately produces innate gross emotions. Simultaneously, stimuli are transmitted from the sensory organs through the cortex of the sensory thalamus to higher-order regions such as the frontal lobe and hippocampus, which process information and transmit the processed information to the amygdala, producing micro-emotions [20].

(2) The role of the auditory system in emotion induction

When responding to external stimuli, individuals automatically develop

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emotional responses. These responses are manifested through physiological and behavioral responses, and these responses vary from situation to situation. The physiological material manifestation of the emotion-inducing mechanism is shown in Figure 2.

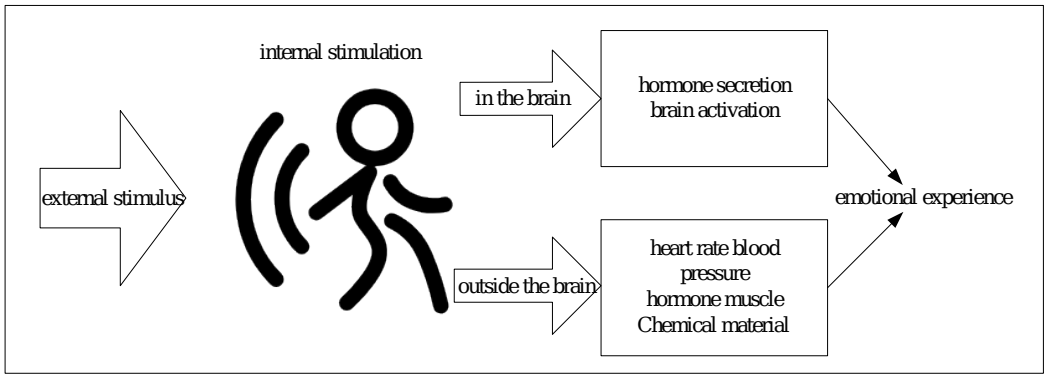

Figure 2. Physiological manifestations of emotion-inducing mechanisms

As shown in Figure 2, in contrast, vocal induction is typically performed using the International Affective Digital System (IADS). An Affective Digital System is a technology or software designed to recognize, interpret, and respond to human emotions and emotional cues. These systems employ various sensors, algorithms, and artificial intelligence to detect and analyze emotional states in individuals and, in some cases, even respond with appropriate emotional expressions or actions. Affective digital systems find applications in areas like human-computer interaction, virtual reality, healthcare, marketing, and more, to create more empathetic and responsive digital experiences. They are often used in emotion recognition, sentiment analysis, and affective computing to improve the interaction between humans and technology. In a comparative study on the effects of visual and auditory affective induction, physiological and behavioral affective responses during acoustic and visual induction were used to compare the effects of inducing media [21].

(3) User emotion recognition framework

The user's listening history contributes to the creation of a personalized song list. To achieve real-time emotion recognition, it's imperative to initially discern the emotions associated with each song in the user's playlist. Building upon this concept and integrating the user's historical listening patterns, an emotion recognition algorithm is devised to compute the user's emotional response while listening to music. The resultant emotional states during music playback are categorized into distinct feelings, including healing, relaxation, romance, nostalgia, excitement, loneliness, and tranquility. Figure 3 illustrates the framework for user emotion recognition.

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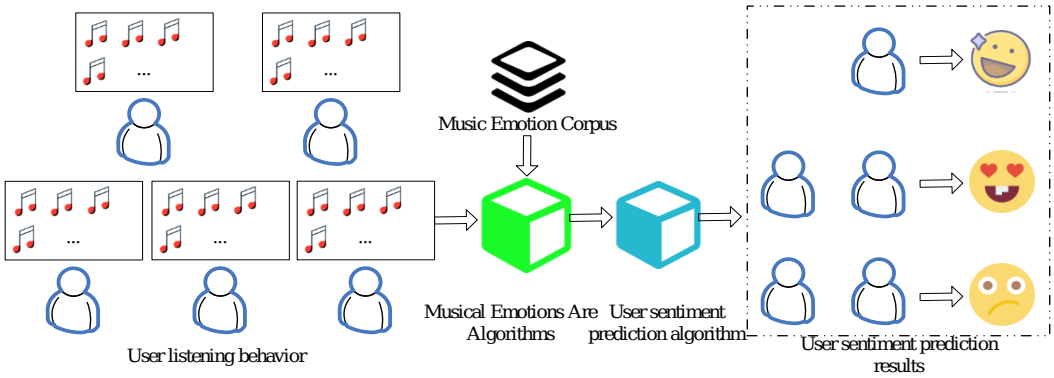

Figure 3. User emotion recognition framework

Figure 3 illustrates a framework comprising three primary components. Initially, it involves the acquisition of historical user data generated during their song-listening activities. Typically, users compile a listening song list after each session, which essentially forms a playlist. It's important to note that the emotional content associated with songs in these playlists may or may not be readily available.

2.2 Sentiment Feature Learning Based on Matrix Factorization

This section will introduce in detail how to learn the emotional representation of songs by using the features of songs and playlists based on the association information between playlists and songs, and the user's emotional annotation information on playlists.

(1) Constrained non-negative matrix factorization

In recent years, the Non-negative Matrix Factorization (NMF) technique has become a popular method for data representation [22]. It appeared to deal with the problem that the input data dimension is too high in the real world. After characterizing the raw data in matrix form, two non-negative matrices can be found by the NMF technique. Their product approximates the original matrix well, thereby simultaneously mapping features in both dimensions into a hidden space [23].

In the context of the identified research gap, Figure 3 presents a framework consisting of three key elements. To address the gap related to cross-cultural emotional recognition and IoT-based systems, there is a crucial need to collect historical data generated by users as they engage in music listening. Users typically create playlists following their listening sessions, which serve as valuable sources of data. It's essential to recognize that the emotional attributes of songs within these playlists may not always be well-documented, and exploring cross-cultural variations in emotion recognition can help enhance the understanding and adaptation of IoT-based emotional recognition systems to diverse cultural contexts.

$$\begin{pmatrix} \square_{11} & \cdots & \square_{1n} \\ \square_{\cdots} & \cdots & \square_{\cdots} \\ \square_{m1} & \cdots & \square_{mn} \end{pmatrix} \tag{1}$$

The goal of non-negative matrix factorization (NMF) is to approximate the original non-negative data matrix X as the product of two lower-dimensional non-

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negative matrices U and V. Formally, the optimization problem can be expressed as:

$$\min_{U,V} O = \|X - UV^T\|_F^2 \tag{2}$$

subject to $U \geq 0$, $V \geq 0$, where $X \in R^{m\times n}$ is the original data matrix, $U \in R^{m\times k}$ is the non-negative basis (feature) matrix, $V \in R^{n\times k}$ is the non-negative coefficient (encoding) matrix, k is the number of latent components (emotional dimensions), and $\|\cdot\|_F$ denotes the Frobenius norm. The constraints $U \geq 0$ and $V \geq 0$ are applied element-wise to ensure interpretability of the factorization.

As shown in Equation (2), minimizing O reduces the reconstruction error between X and its low-rank approximation UV^T, thereby learning an interpretable representation of the data in the latent emotional space.

Since there is no orthogonal constraint here, what we end up with is a r epresentation of the distribution of playlists and songs in emotional space [2 4].

(2) Constrained non-negative matrix factorization incorporating external information

In recent years, many researchers have proposed many methods to incorporate external information into NMF to improve the effect of matrix factorization. This external information is features related to matrix rows or columns [25]. External information is usually incorporated into the objective function of NMF in the form of a regularization term.

1) Emotional label information

Regarding the emotional depiction of Western music, expert-labeled emotional data for song lists or individual songs is integrated into the matrix decomposition as external information. To illustrate this, let's delve into an example using relevant formulas that involve the inclusion of expert-tagged song information. This injection process mirrors the approach of variable V, where the relevant U component is substituted to accommodate the external emotional label data.

$$G_v\|V - V_0\|_F^2 \tag{3}$$

where V is the coefficient matrix from Equation (2), V_0 is the emotional label matrix, and G_v is a diagonal indicator matrix with $G_v(i,i) = 1$ if the ith song has label data and 0 otherwise. As shown in Equation (3), this regularization term constrains V to align with known emotional labels when available, with $\|\cdot\|_F$ denoting the Frobenius norm.

Suppose the sentiment representations learned by matrix factorization are inconsistent with sentiment indications contained in the data. In that case, this loss function will penalize, thereby optimizing the direction of the next step in matrix factorization [26].

2) Relationship network

The adjacency matrix of a graph can be defined as:

$$W^u(i,j) = \begin{cases} 1, \text{ if } u_i \in N(u_i) \text{ or } u_j \in N(u_i) \\ 0, \quad \text{otherwise} \end{cases} \tag{4}$$

u_i is a song in the playlist and $N(u_j)$ is the k nearest neighbors of song j, as

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shown in Equation (4). The neighbors here can be obtained by many metrics and further built into the adjacency matrix. The loss function of the playlist relationship network can be defined as follows:

$$\begin{aligned} R_u &= \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}||U(i,*) - U(j,*)||_2^2 W_u(i,j) \\ &= T_r(U^T(D^u - W^u)U) \\ &= T_r(U^T L^u U) \end{aligned} \tag{5}$$

Among them, $T_r()$ is the trace of the matrix, $L^u = D^u - W^u$ is the Laplacian matrix of the constructed song list graph network, D^u is a diagonal matrix, and $D_u(i,i) = \sum_{j=1}^{m} W^u(i,j)$. If two playlists are close in the graph but have different sentiment labels, this loss function will penalize, as presented in Equation (5). The relationship network definition of the song dimension is similar to that of playlists, replacing the corresponding U with V.

$$sim(i,j) = \frac{V_i V_j}{||V_i||V_j||} \tag{6}$$

$sim(i,j)$ represents the similarity between playlist i and playlist j, and V_i and V_j represent the vectorized representation of the playlist i and playlist j in the feature space, respectively. As shown in Equation (6).

Based on song co-occurrence: A playlist contains multiple songs at the same time, and the same song may also be included in multiple playlists. Playlist similarity is measured by calculating the number of co-occurring songs in different playlists. The more songs that appear in the playlist at the same time, the closer the two playlists are.

Finally, the objective function of the constrained non-negative matrix factorization algorithm incorporating external information can be expressed as:

$$\begin{aligned} \min J = &||X - UV^T||_F^T + \alpha_l^u||G^u(U - U_0)||_F^2 + \alpha_l^v||G^v(V - V_0)||_F^2 + \alpha_c^u Tr(U^T L^u U) \\ &+ \alpha_c^v Tr(V^T L^v V) \end{aligned} \tag{7}$$

Considering the actual meaning of data representation, negative numbers cannot appear in matrices U and V.

(3) Western music emotional representation learning

Let the emotional distribution of the ith song be i, then the song i in the k-dimensional emotional space is expressed as:

$$V_i = (V_{i1}, V_{i2}, \ldots, V_{ik}) \tag{8}$$

The V corresponding to the ith song is normalized, and the emotional probability distribution s_i^* corresponding to the song i is obtained after normalization. The calculation formula is as follows:

$$s_i^* = \left(\frac{V_{ij}}{\sum_{j=1}^{k} V_{ij}}\right) \tag{9}$$

Among them, k represents the emotional space dimension.

For each song, the emotion category with the highest corresponding value is taken as the emotion category to which the song belongs, and the calculation formula is as follows:

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$$e_i^* \leftarrow \mathrm{argmax}(s_i^*) \tag{10}$$

This achieves the purpose of song emotion recognition.

(4) Western music joint representation learning based on Encoder-Decoder

In this study, the encoder and decoder of the Seq2Seq are a Recurrent Neural Network (RNN), respectively. In the RNN structure, the input of the ith layer neuron at time m includes not only the output of the $\square - 1$ layer neuron at time t, but also its output at time $\square - 1$. Assuming that the source modality is the audio modality and the target modality is the lyrics modality, the output of the $\square^{th}$ hidden layer is expressed as:

$$h_t = \mathrm{RNN}\left(h_{t-1}, X_t^A\right) \tag{11}$$

All hidden layers of the encoder RNN are spliced together to form the output of the encoder, which is expressed as:

$$\epsilon_{A\to L} = [h_1, h_2, \ldots, h_T] \tag{12}$$

T is the length of the source mode X^A.

The decoder translates the intermediate representation $\epsilon_{A\to L}$ into the target modality. During decoding, the output at time t depends on both the car and all outputs before time t, which is expressed as:

$$p(X^L) = \prod_{t=1}^{T} p(X_t^L | \epsilon_{A\to L}, X_1^L, \ldots, X_{t-1}^L) \tag{13}$$

The model allows the input of variable-length data. During the training process, the training direction is to maximize the conditional probability. The formula is as follows:

$$\hat{X}^L = \mathrm{argmax} p(X^L | X^A) \tag{14}$$

In the process of modal translation and conversion, to ensure that the model learns the joint representation of all modalities, Cycle Consistency Loss is used as the loss function. Let the function of learning the joint representation $\epsilon_{A\to L}$ between X^A and X^L be f_θ, and the cycle consistency loss function decomposes f_θ into two parts: encoder f_{θ_ε} and decoder f_{θ_d}. Among them, the encoder takes the input and outputs a joint vector $\epsilon_{A\leftrightarrow L}$, which is expressed as:

$$\epsilon_{A\leftrightarrow L} = f_{\theta_\varepsilon}(X^A) \tag{15}$$

The decoder takes $\epsilon_{A\to L}$ as input and outputs X^L, which is expressed as:

$$X^L = f_{\theta_d}(\epsilon_{A\to L}) \tag{16}$$

The above process is the "forward translation" between modalities, which is the process of translating the A mode into the L mode. Back translation is the process of restoring the L mode to the A mode after translating the A mode into the L mode, under the influence of the A mode. Assuming that the L mode obtained by forward translation is $\hat{X}^L$, $\hat{X}^L$ and the A mode obtained by restoring is $\hat{X}^A$, the process of restoring translation can be expressed as:

$$\epsilon_{L\to A} = f_{\theta_\varepsilon}(\hat{X}^L) \tag{17}$$

$$\hat{X}^A = f_{\theta_d}(\epsilon_{L\to A}) \tag{18}$$

The direction of model training and optimization is to maximize the translation conditional probability $p(X^L|X^A)$. The loss function is divided into two parts, including the forward translation loss L and the cycle consistency loss L_C, the

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formula is as follows:

$$L_t = E(I_{X^L}(\hat{X}^L, X^L)) \tag{19}$$

$$L_c = E(I_{X^A}(\hat{X}^A, X^A)) \tag{20}$$

The overall loss function of the model is:

$$L = \alpha_t L_t + \alpha_c L_c \tag{21}$$

α_t and α_c are the weight hyperparameters. The specific loss function L used is the mean square error.

(5) Spectral Angle Mapper (SAM)

The Spectral Angle Mapper (SAM) scale is an essential metric used in this study to evaluate the similarity between spectral signatures by measuring the angle between vectors in a multi-dimensional spectral space. SAM is particularly valuable because it is insensitive to changes in illumination and albedo, focusing purely on spectral shape differences, which makes it highly effective for hyperspectral data analysis and classification. By quantifying spectral similarity through angular measurements rather than absolute reflectance values, SAM provides a robust way to discriminate between classes with subtle spectral differences, thereby enhancing the accuracy and reliability of classification results in our experiments.

3. Experiment on the Effect of Western Music on Emotion Induction

3.1 Experimental Design of Emotion Induction

Electroencephalography (EEG) was employed in this study as it offers a non-invasive, real-time method to capture the brain's electrical activity with high temporal resolution, enabling precise detection of rapid neural responses to musical stimuli and providing objective physiological indicators of emotional states. In this experiment, subjects were asked to listen to Western music clips with three different emotions and to induce an EEG in three different emotional states. The power spectral density information under different rhythms of EEG is a commonly used indicator for EEG analysis. In this experiment, the state of evoked EEG will be classified based on the power spectral features of different EEG rhythms. By comparing the classification accuracy, it is concluded which rhythm power spectral density information is more suitable for the emotion recognition problem. By using different classifiers for the same kind of features and comparing the classification results, the most suitable classifier for the EEG emotion recognition problem is deduced. All the participants were informed about the experiment through verbal informed consent before it began. The institutional review board (IRB) allowed this procedure to proceed since verbal consent could be given in a study where the risk was minimal, as was the case in a non-invasive EEG recording. No personal sensitive or identifiable data were gathered, and participants were completely informed of the procedures involved in the study, their rights, as well as the voluntary nature of the study. The research team recorded the verbal consent as per the IRB guidelines.

(1) Select the target emotion

The distribution of target emotion in the pleasure-arousal emotion space is

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shown in Figure 4.

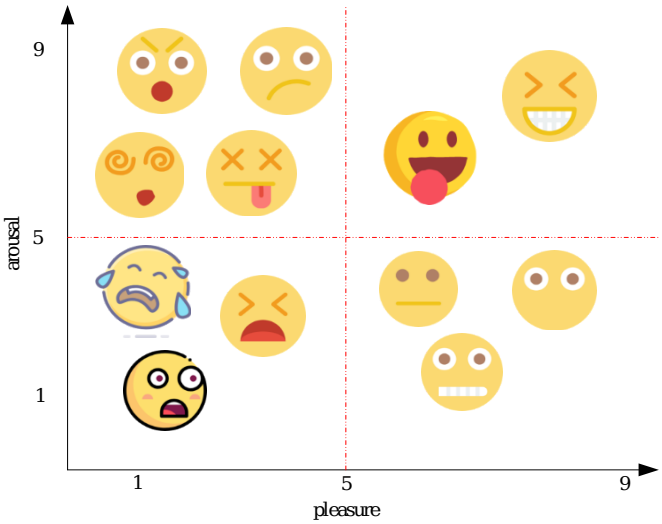
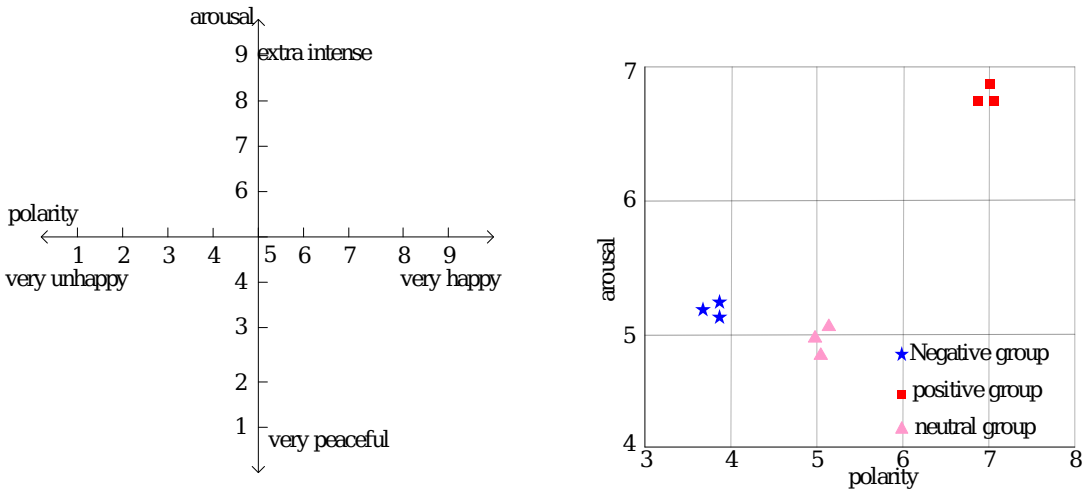


Figure 4. Distribution of target emotion in the pleasure-arousal emotion space

As shown in Figure 4, the pleasure-arousal emotional space is divided into four parts: high pleasure and high arousal Valence=5~9, Arousal=5~9, high pleasure and low arousal Valence=5~9, Arousal =1~5, low pleasure and high arousal Valence=1~5, Arousal=5~9, and low pleasure and low arousal Valence=1~5, Arousal=1~5. Select the most representative emotion type for each space, especially choose sadness and disgust in the space with low pleasure and low arousal. Therefore, the target emotions are composed of: "joy", "disgust", "sorrow", "fear" and "calm".

(2) Measurement of Western Music Emotions

According to a two-dimensional emotion model, emotion can be quantified in two dimensions (polarity and arousal). The two-dimensional model of emotions and the two-dimensional distribution of emotions of experimental stimuli are shown in Figure 5.



(a) Two-dimensional model of emotion

(b) Two-dimensional distribution map of experimental stimulus emotion

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Figure 5. A two-dimensional model of emotion and a two-dimensional distribution map of emotion for experimental stimuli

As shown in Figure 5, different emotions can be represented on a two-dimensional coordinate graph, where the abscissa is polarity and the ordinate is arousal. Polarity represents happiness, and arousal represents calmness or excitement. After listening to Western music, the subjects rated the polarity and arousal, respectively, on an integer ranging from 1 to 9. A score of 1 to 9 on the polarity dimension represents a change from "extremely unpleasant" to "very pleasant." A scale of 1 to 9 on the arousal scale represents a change from "extremely peaceful" to "extremely intense."

(3) Selection of experimental stimuli

132 piano pieces were randomly selected, 7- to 12-second clips were intercepted, and the sound intensity of all clips was normalized to 65dB. Twenty-four college students rated the emotions expressed by these pieces of Western music based on a two-dimensional model of emotions. According to the average score of each Western music segment, 9 Western music segments were selected as experimental stimuli, and the durations of the selected Western music segments were all about 11 seconds. Then, according to the distribution of emotional scores of stimuli on the two-dimensional emotional model, the experimental stimuli were divided into three groups, namely the neutral group, the positive group, and the negative group. The excerpts of Western music that triggered each of the four target emotional categories (Joy, Sadness, Fear, and Disgust) were chosen. The selection was done in two stages. To begin with, we have generated a list of instrumental Western music items, which had already been tested in prior affective neuroscience and music-psychology experiments, and whose emotional valence and arousal qualities were known. The instrumental excerpts were selected in particular in order to prevent linguistic or cultural bias that can be presented by lyrics. Second, an independent group of listeners (not part of the EEG experiment) was also tested in a pilot test to ensure that every excerpt reliably produced the desired emotion. The experiment only included those excerpts that received similar emotional ratings among the participants. This process was done to make sure that every emotional state was a product of a unique and repeatable collection of Western music stimuli. Therefore, the emotions expressed by the Western music clips in the same experimental group can be regarded as the same. And, the polarities of Western musical stimuli were different in the neutral group, the positive group, and the negative group. The polarity of the neutral group is almost 0, the polarity of the positive group is greater than 0, and the polarity of the negative group is less than 0. Paired T-test showed that there were significant differences in polarity between any two experimental groups, so the emotions expressed by Western music in different experimental groups were different.

3.2 Experimental Process

14 healthy subjects who have not received any professional Western music training participated in the experiment. Among them, there were 13 males and 1

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female with an average age of 25.26±2.64 years. Men are right-handed, women are left-handed. The experiment was carried out in a soundproof, shielded room. The subjects sat on a chair 110cm away from the monitor, put on headphones, and were told the basic purpose and procedure of the experiment. The experiment is divided into three groups, and the experimental process is shown in Figure 6.

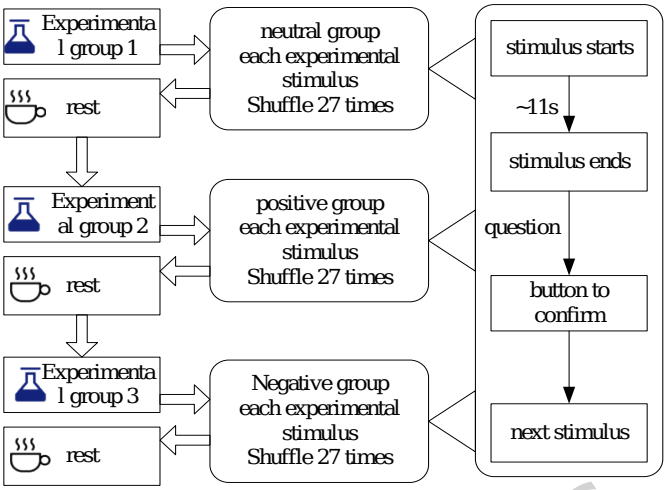

Figure 6. Experimental flow

As shown in Figure 6, the experimental stimuli for the first set of experiments were taken from three pieces of Western music in the neutral group. Before the experiment, the subjects were asked to listen to the three pieces of Western music and memorize their titles, and at the same time, fill in the scale to record the subjects' true feelings about the emotions expressed by the Western music. During the experiment, three Western music stimuli were played randomly and repeated 27 times. The subjects listened to the experimental stimuli with their eyes open and attentive, and tried to avoid head and eye movements. After playing a Western music stimulus, the subject judged the name of the Western music he or she heard by pressing the buttons prompted by the monitor. Adding this button task can make the subjects more focused on completing the experiment. When the subjects completed a keystroke task, they could rest for a while before playing the next Western music stimulus. After the first group of experiments was completed, the subjects rested for a while before starting the second group of experiments. The second set of experimental stimuli came from three Western music clips in the positive group, and the experimental procedure was the same as the previous group. After this group of experiments was completed, the subjects rested for a while, and then began the third group of experiments. The third group of experiments used three pieces of Western music in the negative group as experimental stimuli, and the experimental procedure was the same as the previous group.

Since the duration of the nine Western music stimuli was all around 11 seconds and each stimulus was repeated 27 times, the entire experiment lasted theoretically at least 44.5 minutes. To ensure that the subjects can get enough rest, the actual duration of the experiment varies from 1.5 hours to 2 hours. Since

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negative emotions and positive emotions are evoked by negative and positive Western music, respectively, negative emotions tend to last longer than positive emotions in terms of duration. To avoid the interference of negative emotions on the induction of positive emotions, the three groups of experiments were carried out in the order of first neutral group, then positive group, and finally negative group.

3.3 Emotional Induction Results

In this study, we employ two classification algorithms: the Local Binary Classifier (LBC) and the Local Naive Bayesian Classifier (LNBC). The LBC is a non-parametric classifier that operates by analyzing local binary patterns within the feature space, effectively capturing texture and structural information relevant to the classification task. It is particularly useful for distinguishing subtle variations by encoding local neighborhood relationships. The LNBC, on the other hand, is a probabilistic classifier based on Bayes' theorem, which assumes feature independence and computes the posterior probability of each class. By combining locality principles with the Bayesian framework, the LNBC provides robust classification performance even in scenarios with limited training data. Both classifiers were selected for their complementary strengths in handling the dataset's characteristics and improving overall classification accuracy. The classification results using LBC for subject No. 1's EEG data across five different rhythm characteristics are presented, with similar outcomes observed for other subjects. The x-axis indicates the feature vector dimension, which corresponds to the number of principal components. The blue histogram illustrates the proportion of the total principal components among all components. The red dashed line depicts the trend in the average classification accuracy during cross-validation as a function of feature vector dimension. The error bars along the red dashed line denote the standard deviation of cross-validation classification accuracy. Figures 7 and 8 showcase the classification accuracy obtained through the LBC approach.

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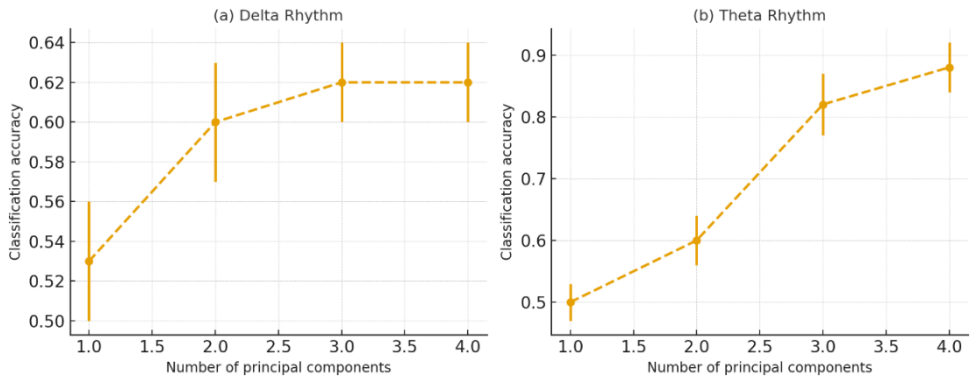

(a) Delta rhythm (b) Theta rhythm

Figure 7. Correct rate of delta rhythm and theta rhythm classification

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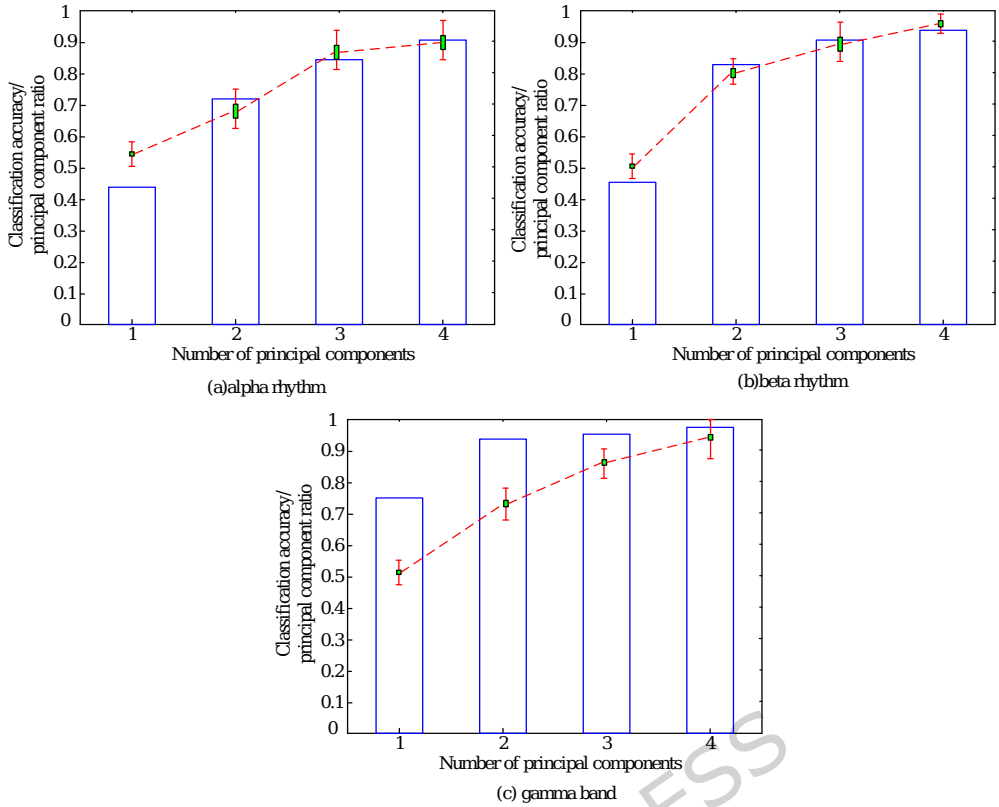

Figure 8. Alpha rhythm, beta rhythm and gamma rhythm classification accuracy

It can be seen from Figures 7 and Figure 8 that the classification accuracy and the proportion of principal components increase with the increase of feature dimensions. When the dimension of the feature vector increases from 3 to 4, the increase in the classification accuracy gradually slows down, and the ratio of the principal components also begins to approach 1. Therefore, if you continue to increase the dimension of the feature vector, not only will the accuracy rate be difficult to improve, but it may also lead to "dimension disaster". Therefore, it is optimal to choose 3 to 4 principal components as feature vectors in this paper. The proportion of people with the highest classification accuracy rate on the five EEG rhythm features is shown in **Table 1**.

Table 1. The proportion of people with the highest classification accuracy rate on the five EEG rhythms

	delta	**theta**	**alpha**	**beta**	**gamma**
LBC	0/14	1/14	0/14	6/14	7/14
LNBC	0/14	1/14	0/14	6/14	7/14
QBC	0/14	3/14	2/14	2/14	7/14
QNBC	0/14	2/14	0/14	8/14	4/14
SVM	0/14	2/14	0/14	4/14	8/14

As shown in **Table 1**, on the delta rhythm, none of the subjects showed the highest classification accuracy, which further indicates that the average power of the delta rhythm is not suitable for studying emotion recognition. Characterized by the average power of theta rhythm or alpha rhythm, a few subjects had the highest classification accuracy. Characterized by the average power of beta or

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gamma rhythms, most subjects obtained the highest classification accuracy.

Figure 9(a) confusion matrix performed with 3 principal components, which illustrates the distribution of the correctly and incorrectly classified samples in the five emotional categories (Joy, Sadness, Fear, Disgust and Calm). Figure 9(b) confusion matrix with the use of 4 principal components, which depicts the better classification results when the diagonal is more dominant. All the cell values are the samples of each of the predicted classes.

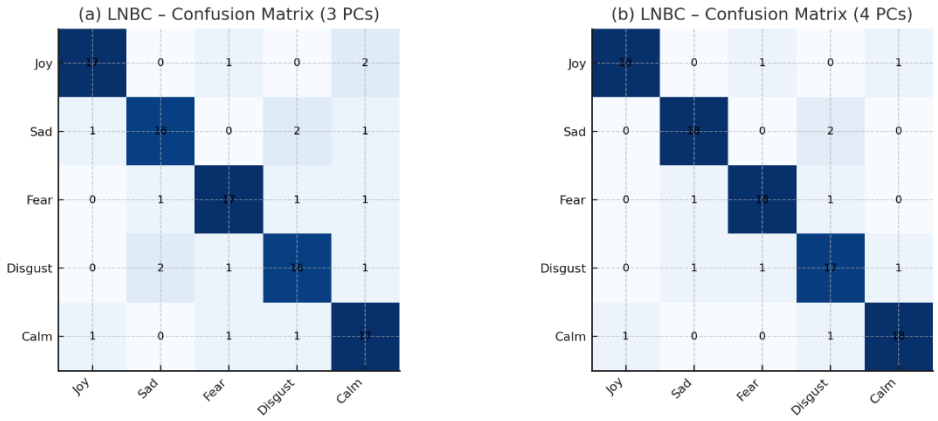

Figure 9. Confusion matrices for the LNBC classifier using different numbers of principal components

Figure 10(a) presents the mean classification error of five EEG rhythms (Delta, Theta, Alpha, Beta, and Gamma). This tendency demonstrates that higher frequency rhythms, especially Beta and Gamma, perform better in decoding with a significantly higher degree than the lower frequency rhythms. Figure 10(b) depicts 95% confidence Intervals of each rhythm have narrower variability in Beta and Gamma, resulting in more consistent and stable classification across subjects. These findings are consistent with the overall observation that high-frequency activity in the cortex provides more discriminatory information for tasks involving emotion recognition.

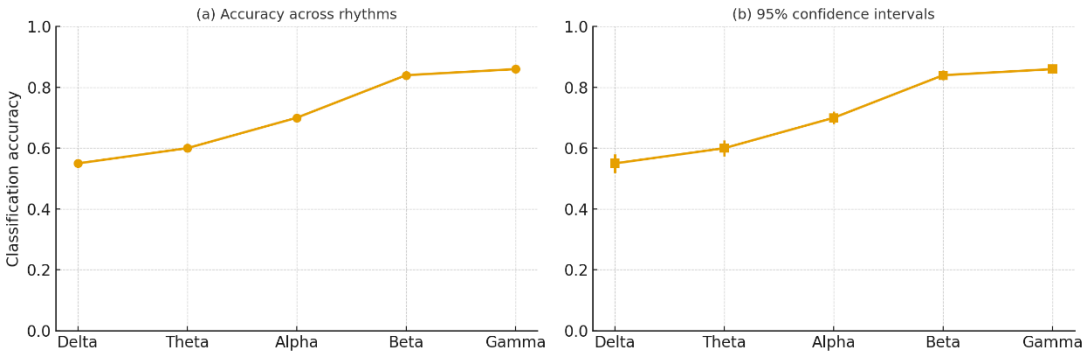

Figure 10. Between-subject classification accuracy comparison across EEG rhythms

Figure 11(a) presents the mean music and video conditions arousal of the heart rate (increase of the baseline in bpm) with the 95% confidence bounds. Music was more likely to cause high physiological arousal among the participants than videos. Figure 11(b) depicts the Self-Assessment Manikin (SAM) arousal ratings of four

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categories of emotions (Joy, Sadness, Fear and Disgust) during music and video conditions. In both instances, music produced slightly higher levels of subjective arousal compared to video, which leads to the finding that Western music produced a stronger emotional activation in the participants.

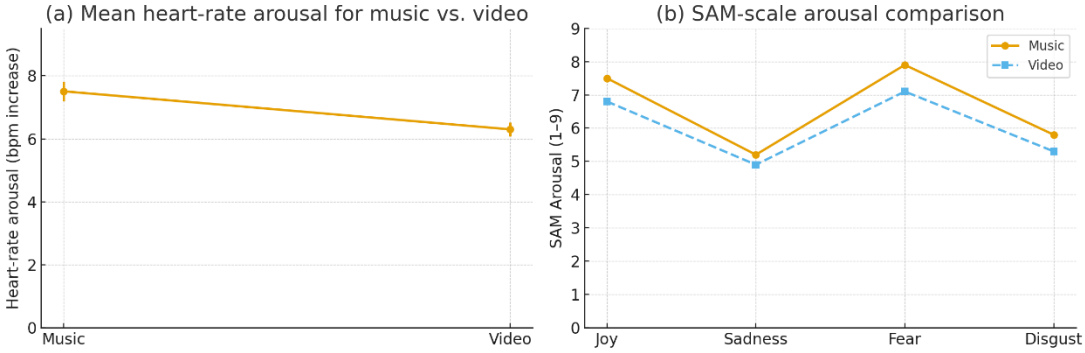

Figure 11. Comparison of emotional arousal induced by music and video stimuli.

First, the blank items were removed, and then the average classification accuracy rate between subjects on each rhythm was calculated separately; the delta rhythm data were not calculated. The average classification accuracy between subjects on the four rhythms is shown in **Tables 2** and **3**.

Table 2. Average between-subject classification accuracy on theta and alpha rhythms

Rhythm	**Theta**		**Alpha**	
Classifier	3	4	3	4
LBC	0.753+0.092	0.829+0.068	0.761+0.101	0.784+0.099
LNBC	0.747+0.112	0.841+0.097	0.743+0.111	0.774+0.131
QBC	0.772+0.112	0.812+0.086	0.745+0.119	0.814+0.115
QNBC	0.722+0.118	0.755+0.082	0.755+0.116	0.756+0.113
SVM	0.795+0.103	0.853+0.111	0.735+0.118	0.799+0.121

Table 3. Average between-subject classification accuracy on beta and gamma rhythms

Rhythm	**Beta**		**Gamma**	
Classifier	3	4	3	4
LBC	0.823+0.099	0.853+0.087	0.785+0.127	0.846+0.113
LNBC	0.838+0.095	0.822+0.083	0.795+0.112	0.831+0.093
QBC	0.836+0.113	0.884+0.121	0.833+0.145	0.901+0.189
QNBC	0.822+0.110	0.837+0.097	0.785+0.134	0.823+0.122
SVM	0.850+0.115	0.882+0.119	0.884+0.082	0.957+0.056

Tables 2 and **3** illustrate that, for each classifier, the average between-subject classification accuracy in beta and gamma rhythms tends to be higher than in theta and alpha rhythms. Regardless of the classifier type and feature vector dimension, the mean between-subject average correctness rates for theta, alpha, beta, and gamma rhythms were 0.787, 0.766, 0.842, and 0.841, respectively. A multiple

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comparison test, at a significance level of 0.05, was conducted on the average classification accuracy rates across subjects in the four rhythms presented in Tables 2 and 3. The results indicate no significant difference between theta and alpha, no significant difference between beta and gamma, but significant differences between either theta or alpha and either beta or gamma in the average correctness rates among subjects. This emphasizes the suitability of average power measurements over beta and gamma rhythms for emotion classification, aligning with previous research findings. A graphical representation of the multiple comparison test for average classification accuracy among subjects under each rhythm is depicted in **Figure 12**.

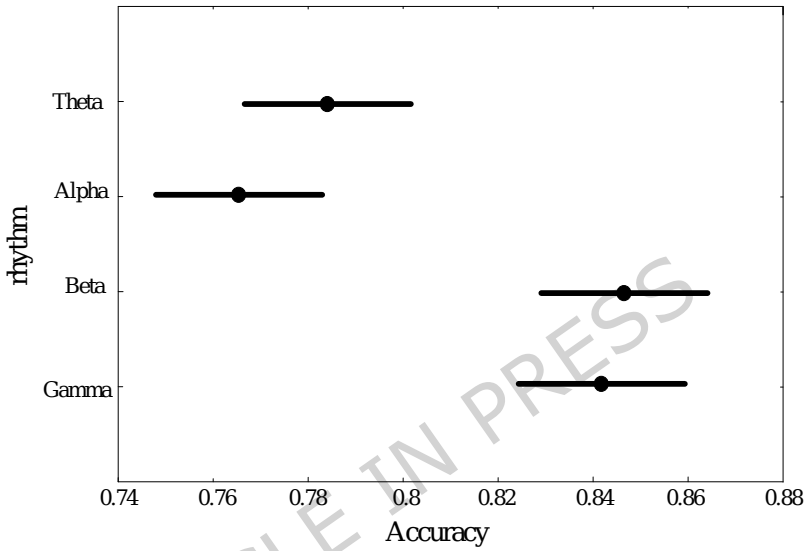


Figure 12. Accuracy vs. Rhythm for the Local Binary Classifier (LBC) and the Local Naive Bayesian Classifier (LNBC).s

As shown in **Figure 12**, the black dot in the figure represents the mean of the average classification accuracy among subjects under a certain rhythm, and the black line represents the 95% confidence interval of the mean. If the confidence intervals of different rhythms overlap, there is no significant difference between the two rhythms, otherwise, there is a significant difference.

Subjects numbered 3, 4, 7, and 13 had blanks on beta and gamma rhythms, because the number of electrodes that could provide classification information was too small to extract enough features for classification. Excluding the above 4 subjects, for the other 10 subjects, multiple comparison tests were used to mark the electrodes with significant differences among the three types of EEGs in the dataset, and the positions of these electrodes that provided classification information were recorded on the topology map.

Figure 13(a) depicts the informative electrode positions of the beta rhythm, indicated on a standard 10 20 head map. The bigger markers represent electrodes that have a greater contribution to classification. Figure 13(b) depicts the distribution of informative electrodes in the gamma rhythm, with more frontal and central sites being involved. These spatial distributions explain the distribution of

the activity with higher frequencies across the cortical areas when processing emotions.

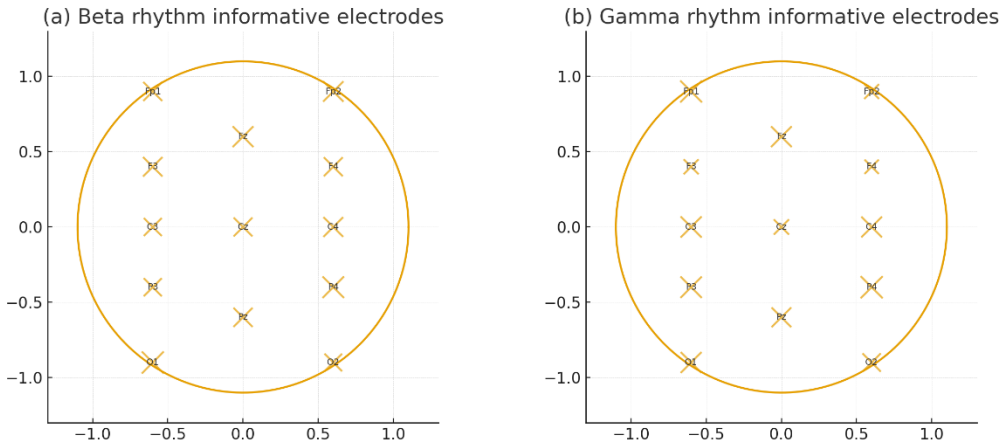
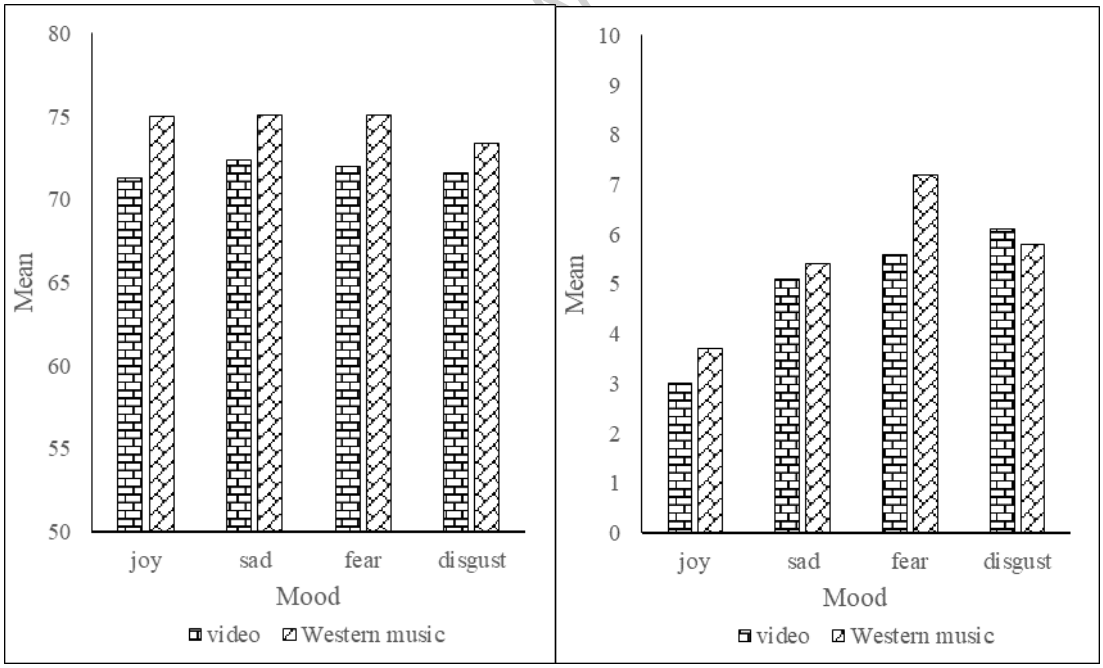


Figure 13. Spatial distribution of informative EEG electrodes for higher-frequency rhythms

All heart rate data were obtained from the Biotrace+ software. The mean value and t-test were used for analysis. The arousal degree of emotion induced by video material and Western music material was counted, and the mean and variance were calculated to calculate the P value. If the P value is less than 0.05, there is a significant difference between the two. Heart rate assessment results. The SAM assessment results are shown in **Figure 14**.



(a) Heart rate assessment results (b) SAM assessment results

Figure 14. Heart rate assessment results, SAM assessment results

As shown in **Figure 14**, the average arousal rate of Western music for the four emotions in terms of heart rate assessment results is higher than that of the video, but the difference is not significant. The results show that the emotional arousal of joy, sadness, fear and disgust in Western music scenes is slightly higher than

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that of video. The results show that the emotional arousal of joy, sadness, fear and disgust in Western music scenes is slightly higher than that of video. According to the statistical analysis of the SAM scale, in the arousal experiment of joy, sadness and fear, the arousal effect of Western music is higher than that of video. And in the arousal of fear, there is a significant difference between the arousal of Western music scenes and the arousal of video. Video arousal was slightly higher in arousal to disgust, but not significantly different. It was observed that increasing the feature vector dimension beyond three or four principal components did not lead to further accuracy improvements; instead, performance plateaued or slightly decreased, likely due to the “curse of dimensionality,” where data sparsity in high-dimensional spaces hampers the classifier’s ability to effectively separate classes. While the present study focused exclusively on Western music to ensure experimental control and dataset consistency, future work will extend the framework to include other musical traditions—such as Chinese classical music—to examine how differences in structural features (e.g., pitch emphasis in Western music versus loudness and rhythm prominence in Chinese music) influence emotional perception and EEG-based classification accuracy.

After the experiment, through feedback with the subjects, the results were consistent with their actual feelings, to analyze the reasons for the better performance of emotionally induced arousal in Western music. On the one hand, it may be affected by the extremely strong negative emotions of fear, and the level of arousal is higher. On the other hand, the real-time panoramic scene of Western music has a strong sense of immersion, a stronger sense of reality, and the plot is more attractive, so it will produce a stronger feeling. During the experiment, in order to improve the accuracy of the experimental data, participants were required not to move their bodies, so the design of Western music scenes removed dynamic interaction. Although this study was conducted within a single cultural context, we recognize that cultural differences can significantly influence emotional perception and musical preference. As part of our future work, we plan to extend this research to include participants from diverse cultural backgrounds, enabling the development of IoT-based emotion recognition models that incorporate culturally specific features and support globally applicable music recommendation systems.

4. Conclusions

The research on automatic emotion recognition has important value, which can make human-computer interaction more humanized. EEG-based emotion recognition has an advantage over other media-based emotion recognition, which can more objectively reflect the subject's internal emotions. This study tried to induce three emotional states through three kinds of emotional music stimuli and collected the corresponding evoked EEGs, and explored the recognition effects of the average power and brain network attribute characteristics under the five EEG rhythms on these three types of EEGs. In emotion recognition based on power spectrum analysis, electrodes that can provide categorical information are first selected by a multiple comparison test. The average power on these electrodes is

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then dimensionally reduced by PCA, and three to four principal components are extracted to form eigenvectors. The classification results show that features on beta and gamma rhythms can provide more effective classification information. There was some inter-subject consistency in the distribution of electrode positions that provided categorical information on beta and gamma rhythms. By comparing the classification effects of different classifiers for the same features, it is found that the performance of SVM and QDC is the best. The study's focus on power spectrum analysis and electrode selection, while informative, highlights the need to explore alternative features and analysis methods for a more comprehensive understanding. The identified inter-subject consistency in electrode positions underscores potential individual variability, and the reliance on specific classifiers may limit the generalizability of classification results. In future work, we plan to expand the scope of our study to include a broader range of emotional states, such as anxiety and excitement, and to incorporate diverse musical genres beyond Western music, thereby enhancing the generalizability and applicability of the proposed music recommendation framework.

Author Contributions Statement:

Conceptualization, D.R.; writing, original draft preparation, D.R.; writing, review and editing, Z.W. All authors have read and agreed to the published version of the manuscript.

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This study has not received any funding.

Ethics approval and consent to participate:

This study was conducted by the ethical standards stipulated in the 1964 Helsinki Declaration, and does not include any animal or human drug experiments. This study obtained verbal informed consent from all participants. After discussion by the Ethics Committee of Zhengzhou University, the project research has been approved (Project Number: 20231002). Before participating in the study, seek verbal informed consent from each participant.

Data Availability Statement:

The EEG datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. Processed feature matrices used for classification, along with associated labels, are included in the public code repository at https://github.com/15776761752-cpu/eeg_emotion_pipeline_complete.git to enable full reproduction of the reported results..

Conflicts of Interest:

We confirmed that there is no conflict of interest.

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